

## QuantERA Proposal Template

Project Acronym

**PA-QML**

Project Title

**Physics-Aware Quantum Machine Learning: From Distributed Hardware to Certified Intelligence**

Topic:



Quantum Phenomena and Resources



Applied Quantum Science

Coordinator contact point for the proposal

|                        |                                                                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                   | Jiook Cha                                                                                                                                              |
| Institution/Department | Seoul National University / Department of Psychology; Department of Brain and Cognitive Sciences; Interdisciplinary Program in Artificial Intelligence |
| Address                | Gwanak-ro 1, Building 16, Suite M512, Gwanakgu, Seoul                                                                                                  |
| Country                | South Korea                                                                                                                                            |
| Phone                  | +82-2-880-8618                                                                                                                                         |
| E-mail                 | connectome@snu.ac.kr                                                                                                                                   |

People involved in the realisation of the project (**please enter these names and their email addresses in the financial template as well**)

| <b>Partner Number</b>          | <b>Country</b> | <b>Institution/Department</b>                                                                                                                          | <b>Name of the Principal Investigator (PI)<sup>1</sup></b> | <b>Name of the co-Investigators<sup>2</sup></b> | <b>Name of the other personnel participating in the project<sup>3</sup></b>                                                                       |
|--------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b><br><i>Coordinator</i> | South Korea    | Seoul National University / Department of Psychology; Department of Brain and Cognitive Sciences; Interdisciplinary Program in Artificial Intelligence | Jiook Cha                                                  |                                                 | Junghoon Justin Park; Maria Pak; Seung Yun Choi; Eunji Lee; Bo-Gyeom Kim; Heehwan Wang; Sangyoon Bae; Jungwoo Seo; Seungju Lee; Jubin Choi; Danny |

<sup>1</sup> The Principal Investigator (PI) is the point of contact of the partner for the corresponding Research Funding Organisation.

<sup>2</sup> A co-investigator is a known scientist and/or group leader making a substantial contribution to the project.

<sup>3</sup> If the name is for the moment unknown, specify the level of expertise sought (PhD, post-doc, engineer, professor...).

|          |             |                                                             |                        |                       |                                                                                                                |
|----------|-------------|-------------------------------------------------------------|------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------|
|          |             |                                                             |                        |                       | Dongyeop Han; Kyungjin Oh; Ahhyun Lucy Lee; Seok Jin Moon; Sebin Lee; Yujin Yun; Allison Eun Se You; Bohee Lee |
| <b>2</b> | South Korea | Yonsei University / Department of Physics                   | Hwidong Yoo            |                       | Sungwon Kim; Yun Eo; Haeun Jang                                                                                |
| <b>3</b> | Germany     | Fraunhofer / Fraunhofer Institute for Cognitive Systems IKS | Jeanette Miriam Lorenz |                       | Alona Sakhnenko                                                                                                |
| <b>4</b> | Italy       | University of Naples Federico II                            | Giovanni Acampora      | Roberto Schiattarella | Autilia Vitiello; Angela Chiatto                                                                               |

Duration: 36 months

**Summary of the project** (publishable abstract, max. 1/2 page):

Be precise and concise. This summary will be used to select suited reviewers for the proposal.

**Relevance to the topic addressed in the call** (in particular specify here which part of the call text is addressed by your project, max. 1/4 page):

Be precise and concise.

# 1. Excellence (max. 6 pages)

## 1.1 Targeted Breakthrough, Baseline of Knowledge and Skills

### Targeted Breakthroughs

This project targets **four fundamental barriers** preventing Quantum Machine Learning (QML) from achieving practical advantage. Current QML remains confined to small-scale demonstrations reliant on heuristic ansatzes that become untrainable at scale. We shift the paradigm from "heuristic QML" to "**Physics-Aware & Certified QML**"—algorithms designed for NISQ constraints rather than ideal hardware assumptions.

**Breakthrough 1 – Scalability via Multi-Chip Ensembles:** We partition high-dimensional data across  $k$  independent QPUs, each processing locally-correlated inputs [1]. **Selective Entanglement** introduces inter-chip quantum connections only for globally-dependent features. Classical aggregation provides algorithmic error mitigation: independent chip errors average out (variance reduction  $\propto 1/k$ ) without resource-heavy correction codes, establishing a **resource theory for "Collective Quantum Advantage"**. This architecture aligns with emerging modular quantum hardware platforms (e.g., IBM Quantum, IonQ, Pasqal), enabling direct deployment as interconnected QPU systems become available.

**Breakthrough 2 – Trainability via QFF-HQGA:** We integrate **Quantum Forward-Forward (QFF)** [2] with **Hybrid Quantum Genetic Algorithm (HQGA)** [3]. QFF decomposes deep circuits into layers optimizing local goodness objectives, mathematically circumventing barren plateaus [4] while preserving expressivity. HQGA encodes parameters as **quantum chromosomes**—superpositions enabling parallel fitness evaluation—using **entangled crossover** and **quantum elitism**.

**Breakthrough 3 – Temporal Expressibility via Q-SSM:** We develop **Quantum State Space Models (Q-SSM)** [5] integrating three-branch quantum superposition with LSTM-style gating. Quantum circuits access  $2^n$ -dimensional Hilbert space with  $O(n \times l)$  parameters versus  $O(N^2)$  classically. The architecture achieves  **$O(L)$  complexity** through chunked processing, matching classical Mamba/Hydra [6], [7] efficiency while exploiting quantum feature expressivity.

**Breakthrough 4 – Reliability via Fuzzy Quantum Diffusion:** We exploit physical quantum noise as a generative resource [8], [9]. **Fuzzy Quantum Logic** via POVMs [10] provides continuous measurements, enabling **Hardware-Attention U-Net** to learn device-specific noise fingerprints. To curb uncertainty from noisy models further, we design **robustness and safety certification** [11], [12] that meet **industrial-grade robustness** [13] standards, with verified resilience to targeted attacks and untargeted perturbations.

### Consortium Expertise:

| Partner                 | Expertise                                                                | Project Role                              |
|-------------------------|--------------------------------------------------------------------------|-------------------------------------------|
| SNU (Cha)               | Multi-Chip Ensembles, Q-SSM, Quantum Transformers, HQTCN                 | Coordinator; Architecture Lead (WP1, WP2) |
| Naples (Acampora)       | Evolutionary Algorithms, Fuzzy Logic, Quantum Computational Intelligence | Optimization & Genetic Model Lead (WP3)   |
| Fraunhofer IKS (Lorenz) | QML, robustness certification, European QC Benchmarking Committee        | Certification Lead (WP5, WP6)             |

| Partner      | Expertise                                                        | Project Role                 |
|--------------|------------------------------------------------------------------|------------------------------|
| Yonsei (Yoo) | CMS/CERN collaboration, Q-ABCDISCo, TCN-VQC, Quantum ViT for HEP | Domain Validation Lead (WP4) |

### Specific Objectives

| Objective               | Goal                                             | Measure                                                  | Target                                                                                                  | Timeline |
|-------------------------|--------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------|
| <b>O1: Scalability</b>  | Multi-Chip Ensembles with Selective Entanglement | Accuracy retention vs monolithic circuit                 | $\geq 90\%$ accuracy at $2\times$ effective qubit scaling; $< 10\%$ variance across configurations      | M12→M30  |
| <b>O2: Trainability</b> | QFF-HQGA hybrid optimization                     | Convergence in barren plateau regimes; measurement count | Convergence at $> 10$ layers where BP fails; $\geq 50\%$ fewer circuit evaluations vs parameter-shift   | M12→M24  |
| <b>O3: Temporal</b>     | Q-SSM with three-branch superposition            | Memory capacity; complexity scaling                      | $\geq 1.5\times$ memory capacity vs classical SSM; verified $O(L)$ scaling                              | M18→M30  |
| <b>O4: Reliability</b>  | Fuzzy Quantum Diffusion & safety certification   | FID under noise; robustness against perturbations        | $FID \leq 1.5\times$ classical baseline under $10^{-3}$ gate error; Performance degradation $\leq 10\%$ | M18→M36  |
| <b>O5: Validation</b>   | Domain science certification of O1-O4            | HEP/Neuro/Cyber benchmarks                               | Statistical quantum advantage ( $p < 0.05$ ) in $\geq 1$ domain metric                                  | M6→M36   |

### O5 Validation Details:

- **HEP:** Multi-Chip Quantum ViT for calorimeter classification; Q-ABCDISCo ( $\sim 1,372$  parameters) with Distance Correlation penalty for background estimation; TCN-VQC (4 qubits) with Noise2Noise self-supervision for waveform denoising
- **Neuroscience:** Q-SSM and Quantum Transformers for EEG/fMRI spatio-temporal analysis (→O3); QFF-HQGA optimized QAOA for brain network modularity optimization and VQE for active inference modeling (→O2)
- **Cybersecurity:** Fuzzy Quantum Diffusion for anomaly detection via hardware fingerprinting; certified robustness against targeted (e.g. adversarial attacks) and untargeted (e.g. input or device noise) perturbations.

## 1.2 Novelty, Level of Ambition and Foundational Character

### Advance Beyond State-of-the-Art

| Challenge          | SOTA Limitation                             | Our Advance                                                                       |
|--------------------|---------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Scalability</b> | Circuit cutting incurs exponential sampling | <b>Resource theory for Collective Quantum Advantage [1]:</b> ensemble aggregation |

| Challenge           | SOTA Limitation                                                                                                                                     | Our Advance                                                                                                                                                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | overhead; no multi-modal fusion theory exists                                                                                                       | preserves expressivity; partitioning eliminates inter-chip entanglement costs; Selective Entanglement optimizes expressivity-trainability trade-off                                                                                                             |
| <b>Trainability</b> | Gradient-based: barren plateaus at $O(n)$ depth [4]. Gradient-free: slow convergence, high sampling. Trainable circuits often classically simulable | <b>Symbiotic QFF-HQGA</b> : local goodness objectives [2] mathematically avoid BP; quantum chromosomes with entangled crossover [3] navigate non-simulable deep circuit landscapes without gradient computation                                                 |
| <b>Temporal</b>     | Quantum RNNs collapse sequences via pooling; no quantum SSM exists                                                                                  | <b>Novel Q-SSM architecture</b> [5]: measurement-based superposition ( $\alpha, \beta, \gamma \in \mathbb{C}$ ) preserves gradient flow; LSTM gating enables selective memory; chunked processing achieves $O(L)$ complexity matching classical Mamba/Hydra [6] |
| <b>Reliability</b>  | Noise treated as error requiring resource-heavy correction; no QML certification standards                                                          | <b>Noise-as-resource paradigm for QML</b> [8]: Fuzzy POVMs [10] enable hardware-native generative modeling; <b>safety certification protocol</b> [11] with formal guarantees.                                                                                   |

## Level of Ambition

**Foundational, not incremental:** We establish new theoretical frameworks—resource theory for distributed QML, mathematical proof of barren plateau avoidance via local objectives, formal certification metrics—rather than algorithmic refinements.

**"Big Science" Validation:** We validate on datasets with genuine complexity, not toy benchmarks:

- **HEP (LHC/CMS):** Petabyte-scale data with complex non-local correlations. Q-ABCDiCo achieves background estimation with  $\sim 1,372$  parameters (vs  $\sim 10^5$  classical). Quantum ViT targets merged electron classification at  $\Delta R \approx 0.001$  resolution.
- **Neuroscience:** High-dimensional, data-scarce clinical signals [14]. Q-SSM captures long-range temporal patterns in EEG/fMRI. QFF-HQGA enables training of QAOA for brain network modularity optimization and VQE for active inference modelling [15].
- **Cybersecurity:** Dynamic threat landscapes requiring certified robustness. Safety certification provides formal adversarial guarantees unprecedented in QML [16].

**Industrial Certification Standard:** we transform QML from experimental demonstrations to certifiable engineering—a prerequisite for safety-critical deployment in automotive, medical, and financial sectors.

## 1.3 Concept and Methodology

### Overall Concept: Physics-Aware QML Stack

We assume NISQ devices remain fragmented and noisy. Rather than waiting for fault tolerance, we redesign algorithms to exploit these constraints.

**Core hypothesis:** Quantum advantage lies in *feature representation* (Hilbert space expressivity with parameter efficiency), not error-free computation. Our hybrid architectures

extract quantum features via shallow, noise-resilient circuits while delegating temporal memory and output generation to classical components proven effective for these tasks.

## Methodology

### M1: Multi-Chip Ensembles (→O1: Scalability)

**Mechanism:** Partition input  $x \in \mathbb{R}^d$  into  $k$  subsets  $\{x_1, \dots, x_k\}$  assigned to independent  $n$ -qubit chips. Each chip  $i$  applies modality-specific ansatz  $U_i(\theta_i)$ . **Selective Entanglement** introduces controlled inter-chip operations only between features with high classical correlation (identified via mutual information analysis). Outputs aggregate via learned weights  $\hat{y} = \sum_i w_i \cdot f(\langle \psi_i | M | \psi_i \rangle)$ .

**Scientific basis:** Partitioning restricts entanglement entropy growth across the full system, provably mitigating barren plateaus [4] (concentration of measure scales with entangled subsystem size, not total qubits). Independent chip errors are uncorrelated; aggregation reduces variance by factor  $1/k$  without correction overhead [1].

### M2: QFF-HQGA Optimization (→O2: Trainability)

**Mechanism:** Decompose  $L$ -layer circuit into  $L$  independent optimization problems. Layer  $l$  minimizes local goodness  $G_l(\theta_l) = \langle \psi_{pos} | M_l | \psi_{pos} \rangle - \langle \psi_{neg} | M_l | \psi_{neg} \rangle$ , where  $\psi_{pos}/\psi_{neg}$  are states from positive (real) and negative (corrupted) data passes. Within each layer, HQGA encodes parameters as quantum chromosomes  $|\theta\rangle = \sum_j \alpha_j |\theta_j\rangle$ . **Entangled crossover** (controlled rotations correlating successful parameter traits) and **quantum elitism** (amplitude amplification of high-fitness states) evolve the population.

**Scientific basis:** Local objectives have  $O(1)$  gradient variance—versus exponentially vanishing global gradients [4]—because each layer's loss landscape is shallow [3]. Quantum parallelism evaluates population fitness in  $O(1)$  circuit executions per generation, eliminating the measurement overhead of parameter-shift gradient estimation.

### M3: Q-SSM Architecture (→O3: Temporal Expressibility)

**Mechanism:** For sequence  $\{x_t\}$ , partition into chunks. Each chunk passes through three parallel VQCs producing measurements  $\{m_\alpha, m_\beta, m_\gamma\}$ . Combine via trainable complex coefficients:

$$h_q = \alpha \cdot m_\alpha + \beta \cdot m_\beta + \gamma \cdot m_\gamma, \text{ where } \alpha, \beta, \gamma \in \mathbb{C}$$

This **measurement-based superposition** enables interference-like feature combination while maintaining gradient flow. Classical LSTM gates process temporal dynamics:

$$c_t = f_t \odot c_{t-1} + i_t \odot h_q o_t = \tanh(c_t) \odot \sigma(W_o \cdot h_q)$$

Chunked processing divides long sequences into manageable segments, with LSTM gates maintaining state across chunks—achieving  $O(L)$  complexity analogous to classical Mamba/Hydra [6], [7] while leveraging quantum Hilbert space expressivity.

**Scientific basis:** Quantum circuits provide  $2^n$ -dimensional feature space with  $O(n \times l)$  parameters versus  $O(N^2)$  for equivalent classical layers. LSTM gating—proven optimal for temporal credit assignment—handles sequence memory. Complexity is  $O(L)$  in sequence length (constant quantum cost per chunk).

### M4: Fuzzy Quantum Diffusion + Reliability & Safety Certification (→O4: Reliability)

**Mechanism:** Forward diffusion uses physical noise channels (amplitude damping, dephasing, crosstalk) instead of synthetic Gaussian noise. **POVMs** provide continuous "fuzzy" measurements  $x_t = \text{Tr}[E_t(\rho) \cdot E_{\text{detect}}]$  where  $E_t$  parameterizes measurement sharpness by decoherence time  $t$ . **Hardware-Attention U-Net** learns device-specific reversal using calibration maps ( $T_1, T_2$ , readout error rates). **Certification** computes:

- **Lipschitz constant:**  $L = \max_{\{x, x'\}} \frac{||f(x)-f(x')||}{||x-x'||}$  (certified robustness bound)

**Scientific basis:** Physical noise is deterministic given device characterization—unlike stochastic Gaussian—enabling precise reversal learning [8]. Fuzzy measurements [10] preserve information lost in sharp projection, maintaining gradient flow for hybrid training. Lipschitz bounds [11] provide formal guarantees: output perturbation ≤ L × input perturbation.

**M5: Domain Validation (→O5)**

| Domain | Task                                  | Method                                        | Quantum Advantage Hypothesis                       |
|--------|---------------------------------------|-----------------------------------------------|----------------------------------------------------|
| HEP    | Background estimation                 | Q-ABCDisCo + DisCo penalty                    | Parameter efficiency (~1,372 vs ~10 <sup>5</sup> ) |
|        | Waveform denoising                    | TCN-VQC (4 qubits) + Noise2Noise              | Self-supervised learning without clean labels      |
|        | Merged electron ID                    | Multi-Chip Quantum ViT                        | ΔR ≈ 0.001 resolution via quantum attention        |
| Neuro  | EEG/fMRI analysis                     | Q-SSM + Quantum Transformers                  | Temporal expressivity (→O3)                        |
|        | Brain network modularity optimization | QFF-HQGA optimized QAOA/VQE                   | Trainability for combinatorial optimization (→O2)  |
| Cyber  | Anomaly detection                     | Fuzzy Quantum Diffusion                       | Hardware fingerprint sensitivity                   |
|        | Robustness certification              | Targeted / untargeted perturbation robustness | Adversarial robustness and data privacy            |

**Risk Mitigation**

| Risk                                        | Likelihood | Mitigation Strategy                                                       |
|---------------------------------------------|------------|---------------------------------------------------------------------------|
| Barren plateaus persist despite QFF         | Medium     | Fallback to pure HQGA (entirely gradient-free); reduce circuit depth      |
| Ensemble accuracy degradation               | Medium     | Weighted voting by per-chip reliability scores; adaptive chip selection   |
| Hardware noise exceeds Fuzzy model capacity | Low        | Pivot to Fuzzy Quantum GANs with stabilized training dynamics             |
| Domain data access delays                   | Low        | CERN Open Data Portal; PhysioNet EEG; public intrusion detection datasets |



## 1.4 Interdisciplinary Nature

### Research Disciplines and Integration

| Discipline                                    | Contribution                                             | Integration Point                                                |
|-----------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------|
| <b>Quantum Information Science</b>            | Ansatz design, entanglement theory, variational circuits | Provides computational substrate and expressivity theory         |
| <b>Computational Intelligence</b>             | Fuzzy Logic, Evolutionary Algorithms                     | Solves trainability/robustness where pure QIS methods fail       |
| <b>Software Engineering &amp; Reliability</b> | benchmarking, certification protocols                    | Transforms theory into deployable, verifiable standards          |
| <b>Domain Sciences (HEP, Neuroscience)</b>    | Complex real-world datasets and domain constraints       | Provides validation complexity; grounds quantum advantage claims |

**Cross-Fertilization:** Annual methodology exchanges ensure knowledge transfer (Naples: Fuzzy/evolutionary methods; SNU: Q-SSM/Multi-Chip protocols; Fraunhofer: Safety certification). Domain Challenge Sprints apply theoretical advances to real problems—HEP jet classification, EEG/fMRI temporal modelling, brain network modularity optimization, adversarial robustness testing—evaluated by domain-specific metrics.

### Interdisciplinary Synergies:

- **Fuzzy Logic + Quantum Mechanics:** Biological uncertainty (neuroscience) and hardware noise (NISQ) unified as continuous "degrees of truth" via POVMs
- **Evolutionary Algorithms + Quantum Circuits:** QML models that *evo/ve* through non-convex landscapes where gradient-based optimization fails
- **Distributed Computing + HEP:** Multi-Chip protocols validated on particle data with intrinsic non-local correlations—ideal for demonstrating ensemble aggregation preserves quantum expressivity

A unified software architecture with standardized interfaces (Physics, Logic, Certification modules) ensures integration at the code level.

## 1.5 Gender Dimension and Open Science Practices

### Gender Dimension

The foundational QML algorithms are gender-neutral. For the **Neuroscience validation (O5)**, where EEG/fMRI data involves human subjects, we implement **gender-stratified validation**: balanced training data (50/50 male/female) and sex as an explicit covariate to prevent algorithmic bias in brain signal analysis.

The consortium includes **female leadership** (PD Dr. habil. Jeanette Miriam Lorenz, Fraunhofer IKS) as Principal Investigator, serving as a role model for early-career researchers.

### Open Science Practices

- **Open Access:** 100% of publications via Green (arXiv, Zenodo) or Gold routes.
- **FAIR Data:** Challenge datasets (anonymized EEG, LHC subsets) deposited in Zenodo with DOIs. Data Management Plan delivered by M3.
- **Open Source:** "Physics-Aware QML" library released under Apache 2.0 license on GitHub, with Jupyter notebook tutorials for reproducibility.
- **Sensitive Data:** Neuroscience data pseudonymized at source (GDPR/PIPA compliant). Cybersecurity logs screened for PII before consortium sharing.

## 2. Impact (max. 3 pages)

### 2.1 Expected Impacts

This project addresses the QuantERA QPR call objective to *"explore novel quantum phenomena and resources... and address major challenges preventing broad applications."* We establish a **resource theory for scalable and reliable QML** with the following impacts:

#### Quantified Impact Indicators

| Impact                | Objective | Indicator                              | Baseline                         | Target                                              | Timeline |
|-----------------------|-----------|----------------------------------------|----------------------------------|-----------------------------------------------------|----------|
| Distributed Computing | O1        | Accuracy retention at 2× qubit scaling | Single-chip: ~20-50 qubits       | ≥90%, variance $\propto 1/k$                        | M30      |
| Trainability          | O2        | Convergence in barren plateau regimes  | Gradient methods fail >10 layers | QFF-HQGA converges >10 layers                       | M24      |
| Temporal Learning     | O3        | Memory capacity vs. classical SSM      | Quantum RNNs collapse sequences  | ≥1.5× classical; $O(L)$ complexity                  | M30      |
| Reliability           | O4        | Certified robustness (Lipschitz bound) | No QML certification exists      | $L < 2.0$ ; FID $\leq 1.5\times$ at $10^{-3}$ error | M36      |

#### Key Contributions:

- **O1:** "Collective Quantum Advantage" via Multi-Chip Ensembles [1]—ensemble aggregation provides algorithmic error mitigation (variance  $\propto 1/k$ ) without correction codes
- **O2:** QFF-HQGA resolves the optimization trilemma—local goodness objectives [2] avoid barren plateaus; quantum evolutionary search [3] eliminates gradient overhead
- **O3:** Q-SSM [5] achieves  $O(L)$  complexity matching classical Mamba [6] while exploiting  $2^n$ -dimensional Hilbert space with  $O(n \times l)$  parameters
- **O4:** Safety certification [11], [12] provides formal Lipschitz guarantees; Fuzzy POVMs [10] enable noise-as-resource paradigm [8]

#### Domain Science Validation (→O5)

| Domain | Task                  | Method                 | Quantum Advantage Hypothesis                | Success Metric          |
|--------|-----------------------|------------------------|---------------------------------------------|-------------------------|
| HEP    | Background estimation | Q-ABCDiCo              | ~1,372 vs. $\sim 10^5$ classical parameters | $p < 0.05$              |
|        | Waveform denoising    | TCN-VQC + Noise2Noise  | Self-supervised without clean labels        | SNR improvement         |
|        | Merged electron ID    | Multi-Chip Quantum ViT | $\Delta R \approx 0.001$ resolution         | Classification accuracy |

| Domain | Task                                  | Method                  | Quantum Advantage Hypothesis                 | Success Metric                    |
|--------|---------------------------------------|-------------------------|----------------------------------------------|-----------------------------------|
| Neuro  | EEG/fMRI analysis                     | Q-SSM                   | Long-range temporal patterns                 | Memory $\geq 1.5\times$ classical |
|        | Brain network modularity optimization | QFF-HQGA QAOA/VQE       | Trainability for combinatorial problems [15] | Convergence                       |
| Cyber  | Anomaly detection                     | Fuzzy Quantum Diffusion | Hardware fingerprint sensitivity             | Detection rate                    |
|        | Robustness certification              | Noise in QML [13]       | Formal adversarial guarantees                | $L < 2.0$                         |

### Transformational Societal Impact

| Impact Area                          | Gap Addressed                                  | Our Contribution                                                      | Societal Benefit                                                             |
|--------------------------------------|------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Industrial Certification</b>      | No QML safety/robustness standards             | Pipeline-integrated certification (Lipschitz bounds, noise stability) | Enables quantum AI in safety-critical sectors (automotive, medical, finance) |
| <b>Democratization</b>               | Quantum advantage restricted to elite labs     | Multi-Chip Ensembles aggregate smaller QPUs                           | Universities/SMEs access "virtual large-scale" QML                           |
| <b>Privacy-Preserving Healthcare</b> | Medical data sharing constrained by GDPR/HIPAA | Fuzzy Quantum Diffusion generates synthetic data [9]                  | Unlocks medical AI development without privacy compromise                    |
| <b>Fundamental Physics</b>           | LHC analysis bottlenecks                       | Quantum ViT/Q-ABCDiCo with superior parameter efficiency              | Accelerates discovery beyond Standard Model                                  |

## 2.2 Dissemination, Exploitation, and Communication

### Dissemination Strategy

- **Publications ( $\geq 12$  peer-reviewed, 100% Open Access):**

| Type         | Venues                               | Timeline | Lead               |
|--------------|--------------------------------------|----------|--------------------|
| Theory       | <i>Nature Comms, PRX Quantum</i>     | M12-M24  | SNU, AII           |
| Methods      | <i>npj Quantum Info, QST</i>         | M18-M30  | SNU, Naples        |
| Applications | <i>PRL, NeuroImage, IEEE S&amp;P</i> | M24-M36  | Yonsei, Fraunhofer |
| Tools        | <i>Quantum Machine Intelligence</i>  | M30-M36  | Fraunhofer, Naples |

- **Conferences:** IEEE QAI, IEEE QCE (2-3 papers each); NeurIPS/ICML workshops; domain conferences (CHEP, OHBM, IEEE S&P)
- **Training:** Annual consortium workshops (M12, M24, M36); Joint summer school "Practical QML for Domain Scientists" (M18)

## Exploitation Plan

| Innovation                                     | IP Strategy                | Pathway                      |
|------------------------------------------------|----------------------------|------------------------------|
| Multi-Chip Ensemble Protocol                   | Open-source (Apache 2.0)   | PennyLane/Qiskit integration |
| QFF-HQGA Optimizer                             | Open-source                | Community adoption           |
| Certification Extensions to Software Pipelines | Joint IP (Fraunhofer)      | Industrial licensing         |
| Domain Models & Datasets                       | Open (Zenodo, HuggingFace) | Benchmark standardization    |

- **Software Deliverables:** QML-Ensemble Library (GitHub); Certification Module; Domain Benchmark Suites (Q-ABCDISCo, Q-SSM-EEG, Fuzzy-QIDS)
- **Commercialization:** Hardware vendors (IBM, IQM, Pasqal)—Multi-Chip protocols; Pharma/Biotech—synthetic data licensing; Automotive/Aerospace—Fraunhofer certification consulting
- **Standardization:** IEEE P7131 (Quantum Performance Metrics); QED-C benchmarking; ISO/IEC JTC 1/SC 42 (AI certification)

## Communication Activities

| Audience     | Channels                                | KPIs                           |
|--------------|-----------------------------------------|--------------------------------|
| Scientific   | Publications, GitHub, conferences       | 500+ GitHub stars, 100+ forks  |
| Industry     | Workshops, white papers, advisory board | 200+ webinar registrations     |
| Policymakers | Policy briefs (M18, M36)                | 50+ decision-makers reached    |
| Public       | Website, social media, video series     | 5,000+ visitors, 10,000+ views |

## Data Management (FAIR Principles)

| Data Type          | Repository    | Access     | Timeline   |
|--------------------|---------------|------------|------------|
| Benchmark datasets | Zenodo (DOI)  | Open       | M24        |
| Trained models     | HuggingFace   | Open       | M30        |
| Source code        | GitHub        | Apache 2.0 | Continuous |
| Raw data           | Institutional | Restricted | M36        |

**Reproducibility:** Docker containers, Jupyter notebooks, OSF pre-registration for validation experiments.

### 3. Implementation

#### 3.1 Work Plan (max. 2 pages)

##### Overall Structure

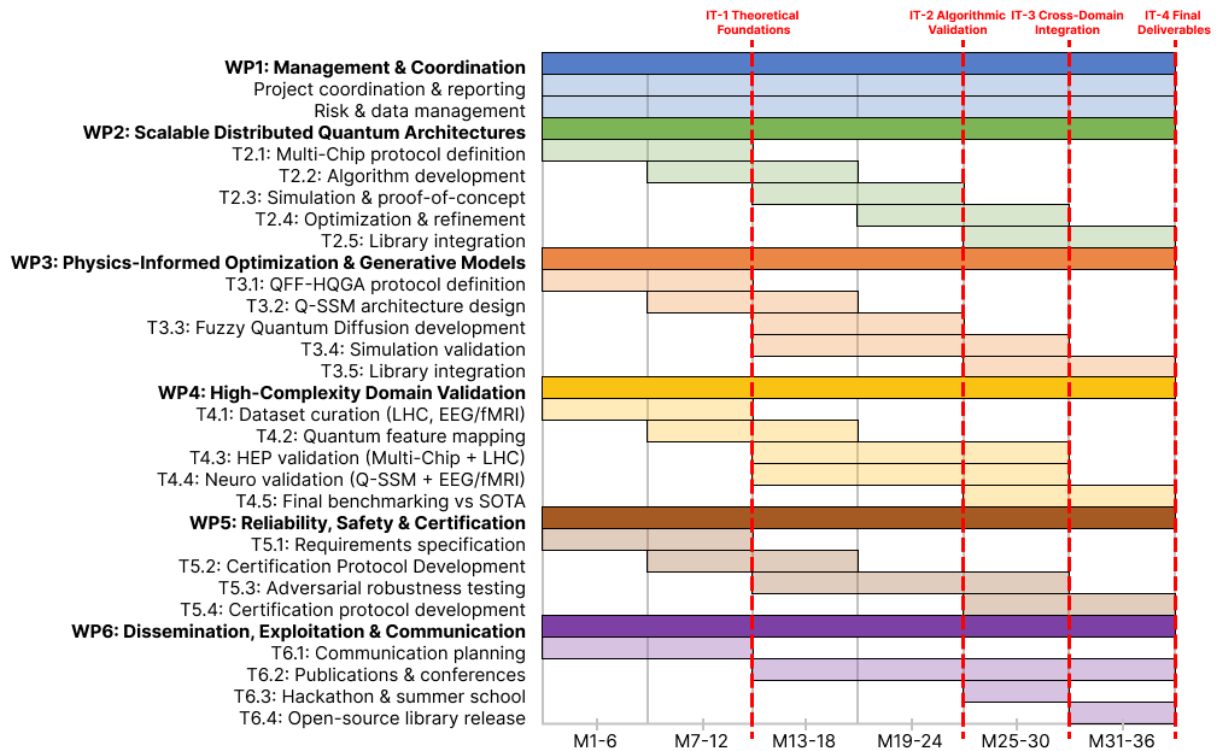
The project is organized into **six interconnected Work Packages (WPs)** creating a feedback loop between foundational theory development and empirical validation:

| WP  | Title                                             | Lead       | Focus                                                      |
|-----|---------------------------------------------------|------------|------------------------------------------------------------|
| WP1 | Management & Coordination                         | SNU        | Scientific alignment, reporting, risk/data management      |
| WP2 | Scalable Distributed Quantum Architectures        | SNU        | Multi-Chip Ensemble protocols (→O1)                        |
| WP3 | Physics-Informed Optimization & Generative Models | Naples     | QFF-HQGA, Q-SSM, and Fuzzy Quantum Diffusion (→O2, O3, O4) |
| WP4 | High-Complexity Domain Validation                 | Yonsei     | HEP and Neuroscience benchmarking (→O5)                    |
| WP5 | Reliability, Safety & Certification               | Fraunhofer | Robustness certification (→O4)                             |
| WP6 | Dissemination, Exploitation & Communication       | Fraunhofer | Open Science, software release, outreach                   |

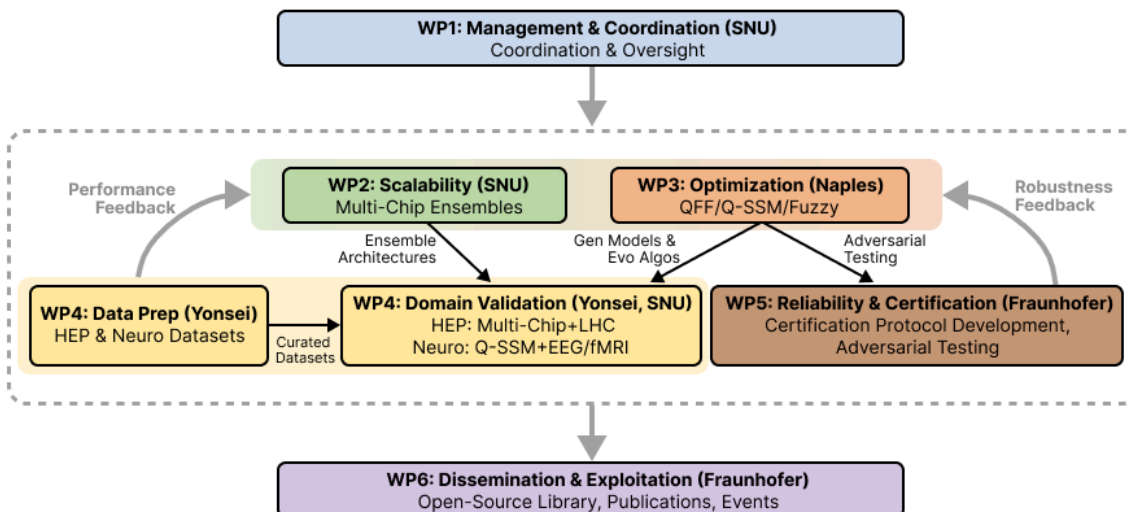
##### Intermediate Targets

| Target | Month | Description                         | Success Criteria                                                                                                                                                                                            |
|--------|-------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IT-1   | M12   | Theoretical Foundations Established | Multi-Chip protocol specs complete; Q-SSM architecture defined; Fuzzy Logic Controllers specified; Domain datasets preprocessed                                                                             |
| IT-2   | M24   | Algorithmic Validation (Simulation) | QFF-HQGA converges on >10-layer circuits where backpropagation fails; Q-SSM achieves $\geq 1.5\times$ classical memory capacity; Fuzzy-Diffusion achieves FID $\leq 1.5\times$ classical diffusion baseline |
| IT-3   | M30   | Cross-Domain Integration            | Multi-Chip Ensemble reconstructs LHC particle correlations; Cybersecurity models pass robustness certification                                                                                              |
| IT-4   | M36   | Final Deliverables                  | Open-source "Physics-Aware QML" library released; Final benchmarks published                                                                                                                                |

**Gantt Chart: Work Package Timeline**



**PERT Diagram: Work Package Dependencies**



**Key Dependencies:**

- **WP2 → WP4:** Multi-Chip Ensemble protocols feed into HEP scalability validation
- **WP3 → WP4:** Q-SSM feeds into Neuro spatio-temporal analysis; QFF-HQGA feeds into brain network modularity optimization; Fuzzy-Diffusion feeds into Cyber anomaly detection
- **WP3 → WP5:** Evolutionary QML models undergo adversarial robustness testing
- **WP4/WP5 → WP2/WP3:** Validation feedback refines foundational algorithms
- **All → WP6:** Validated results flow to dissemination

## 3.2 Work Packages

|                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                 |                   |                          |               |                   |                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------|---------------|-------------------|------------------|
| WP 1                                                                                                                                                                                                  | Project Management & Coordination                                                                                                                                                                                                                                               |                   |                          |               | Start<br>Month: 1 | End<br>Month: 36 |
| Contribution of project partners                                                                                                                                                                      |                                                                                                                                                                                                                                                                                 |                   |                          |               |                   |                  |
| Partner number <sup>4</sup>                                                                                                                                                                           |                                                                                                                                                                                                                                                                                 | 1. SNU            | 2. Yonsei                | 3. Fraunhofer | 4. Naples         |                  |
| Total effort per partner<br>(person*months)                                                                                                                                                           |                                                                                                                                                                                                                                                                                 | 9                 | 1                        | 2             | 1                 |                  |
| Aim of the WP                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                 |                   |                          |               |                   |                  |
| To ensure efficient administrative and scientific coordination, timely reporting to national funding agencies and the QuantERA secretariat, and effective risk/data management across the consortium. |                                                                                                                                                                                                                                                                                 |                   |                          |               |                   |                  |
| Tasks                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                 |                   |                          |               |                   |                  |
| T1.1                                                                                                                                                                                                  | Consortium Coordination & Monitoring (M1–M36; SNU; All) <sup>5</sup><br>Day-to-day management of the project; organization of bi-annual consortium meetings (virtual/physical); monitoring of milestones and KPIs; preparation of periodic progress reports.                    |                   |                          |               |                   |                  |
| T1.2                                                                                                                                                                                                  | Risk & Innovation Management (M1–M36; SNU; Fraunhofer)<br>Continuous monitoring of scientific and technical risks; execution of mitigation plans; management of Intellectual Property Rights (IPR); coordination of innovation activities.                                      |                   |                          |               |                   |                  |
| T1.3                                                                                                                                                                                                  | Data Management & Ethics (M1–M36; SNU; Fraunhofer)<br>Implementation of the Data Management Plan (DMP) ensuring FAIR principles; oversight of ethical compliance regarding EEG/fMRI data (Neuroscience) and sensitive network logs (Cybersecurity); GDPR compliance monitoring. |                   |                          |               |                   |                  |
| Deliverable                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                 | Month of delivery | Title of deliverable     |               |                   |                  |
| D1.1                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                 | M3                | Data Management Plan     |               |                   |                  |
| D1.2                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                 | M18               | Mid-term Progress Report |               |                   |                  |
| D1.3                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                 | M36               | Final Project Report     |               |                   |                  |

<sup>4</sup> **Bold** the partner number of the work package leader

<sup>5</sup> For instance: T1.1 Development of something (M3-M6; responsible: 3; involved: 1, 4)

|                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                          |                   |                                                     |               |                   |                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------|---------------|-------------------|------------------|
| WP 2                                                                                                                                                                                                                                                               | Scalable Distributed Quantum Architectures                                                                                                                                                                                               |                   |                                                     |               | Start<br>Month: 1 | End<br>Month: 36 |
| Contribution of project partners                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                          |                   |                                                     |               |                   |                  |
| Partner number                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                          | 1. SNU            | 2. Yonsei                                           | 3. Fraunhofer | 4. Naples         |                  |
| Total effort per partner (person*months)                                                                                                                                                                                                                           |                                                                                                                                                                                                                                          | 60                | 6                                                   | 4             | 9                 |                  |
| Aim of the WP                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                          |                   |                                                     |               |                   |                  |
| To develop theoretical protocols for "Virtual Large-Scale" quantum computing by implementing Multi-Chip Ensembles, enabling scalable QML on fragmented NISQ hardware via parallel model aggregation. This WP directly addresses <b>Objective 1 (Scalability)</b> . |                                                                                                                                                                                                                                          |                   |                                                     |               |                   |                  |
| Tasks                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                          |                   |                                                     |               |                   |                  |
| T2.1                                                                                                                                                                                                                                                               | Multi-Chip Protocol Definition (M1–M12: SNU; Naples, Fraunhofer)                                                                                                                                                                         |                   |                                                     |               |                   |                  |
|                                                                                                                                                                                                                                                                    | Mathematical formulation of ensemble learning strategies for distributed QPUs; development of aggregation logic (voting/averaging) to combine outputs from independent quantum models; definition of inter-chip communication protocols. |                   |                                                     |               |                   |                  |
| T2.2                                                                                                                                                                                                                                                               | Algorithm Development (M7–M18; SNU; Naples, Fraunhofer)                                                                                                                                                                                  |                   |                                                     |               |                   |                  |
|                                                                                                                                                                                                                                                                    | Design of scalable QML architectures (Quantum Transformers, HQTCN) optimized for ensemble deployment; development of Multi-Modal Data Fusion techniques for heterogeneous data (e.g., sMRI on Chip A, fMRI on Chip B).                   |                   |                                                     |               |                   |                  |
| T2.3                                                                                                                                                                                                                                                               | Simulation & Proof-of-Concept (M13–M24; SNU; Naples, Fraunhofer)                                                                                                                                                                         |                   |                                                     |               |                   |                  |
|                                                                                                                                                                                                                                                                    | Implementation of Multi-Chip protocols in simulation environments; validation of scalability claims on >20 qubit distributed systems; verification of barren plateau mitigation through restricted global entanglement.                  |                   |                                                     |               |                   |                  |
| T2.4                                                                                                                                                                                                                                                               | Optimization & Refinement (M19–M30; SNU; Fraunhofer)                                                                                                                                                                                     |                   |                                                     |               |                   |                  |
|                                                                                                                                                                                                                                                                    | Optimization of qubit-per-chip requirements; refinement based on feedback from WP4 (HEP validation) and WP5 (reliability testing); performance tuning for real-world data characteristics.                                               |                   |                                                     |               |                   |                  |
| T2.5                                                                                                                                                                                                                                                               | Library Integration (M25–M36; SNU; Naples; Fraunhofer)                                                                                                                                                                                   |                   |                                                     |               |                   |                  |
|                                                                                                                                                                                                                                                                    | Integration of validated Multi-Chip Ensemble protocols into the unified "Physics-Aware QML" software library; documentation and API standardization.                                                                                     |                   |                                                     |               |                   |                  |
| Deliverable                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                          | Month of delivery | Title of deliverable                                |               |                   |                  |
| D2.1                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                          | M12               | Report on Multi-Chip Protocols                      |               |                   |                  |
| D2.2                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                          | M24               | Software Module for Distributed Circuit Aggregation |               |                   |                  |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                 |                   |                                                                               |                |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------|----------------|---------------|
| WP 3                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Physics-Informed Optimization & Generative Models                                                                                                                                                                                                                                                                                                               |                   |                                                                               | Start month: 1 | End month: 36 |
| Contribution of project partners                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                 |                   |                                                                               |                |               |
| Partner number                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1. SNU                                                                                                                                                                                                                                                                                                                                                          | 2. Yonsei         | 3. Fraunhofer                                                                 | 4. Naples      |               |
| Total effort per partner (person*months)                                                                                                                                                                                                                                                                                                                                                                                                             | 36                                                                                                                                                                                                                                                                                                                                                              | 2                 | 4                                                                             | 24             |               |
| <b>Aim of the WP</b><br>To solve the "Trainability" and "Noise" bottlenecks by developing the Quantum Forward-Forward algorithm (QFF) driven by Hybrid Quantum Genetic Algorithms (HQGA), Quantum State Space Models (Q-SSM) for temporal expressibility, and Fuzzy-controlled Quantum Diffusion models. This WP addresses <b>Objective 2 (Trainability)</b> , <b>Objective 3 (Temporal Expressibility)</b> , and <b>Objective 4 (Reliability)</b> . |                                                                                                                                                                                                                                                                                                                                                                 |                   |                                                                               |                |               |
| Tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                 |                   |                                                                               |                |               |
| T3.1                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>QFF-HQGA Protocol Definition (M1–M12; Naples; SNU)</b><br>Development of Quantum Forward-Forward (QFF) algorithm with local "goodness" objectives; integration with Hybrid Quantum Genetic Algorithms (HQGA) for gradient-free optimization; benchmarking convergence against Backpropagation in Barren Plateau landscapes.                                  |                   |                                                                               |                |               |
| T3.2                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Q-SSM Architecture Design (M7–M18; SNU; Naples)</b><br>Development of Quantum State Space Models (Q-SSM) integrating variational quantum circuits with classical LSTM-style gating mechanisms; implementation of three-branch quantum superposition for measurement-based interference effects; design of chunked recurrence processing for sequential data. |                   |                                                                               |                |               |
| T3.3                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Fuzzy Quantum Diffusion Development (M13–M24; Naples; SNU)</b><br>Design of Fuzzy Logic Controllers (FLC) to model NISQ noise as uncertainty; integration of FLC with Positive Operator-Valued Measures (POVMs) for continuous "unsharp" measurements; development of Hardware-Attention U-Net for reverse diffusion process.                                |                   |                                                                               |                |               |
| T3.4                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Simulation Validation (M13–M30; Naples; SNU, Fraunhofer)</b><br>Full implementation of QFF-HQGA, Q-SSM, and Fuzzy-Quantum Diffusion models; simulation-based validation of trainability improvements and noise resilience; comparison with classical baselines.                                                                                              |                   |                                                                               |                |               |
| T3.5                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Library Integration (M25–M36; Naples; SNU, Fraunhofer)</b><br>Integration of all optimization and generative model algorithms into the unified "Physics-Aware QML" library; API development and documentation.                                                                                                                                               |                   |                                                                               |                |               |
| Deliverable                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                 | Month of delivery | Title of deliverable                                                          |                |               |
| D3.1                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                 | M18               | Algorithm Definition: QFF-HQGA, Q-SSM & Fuzzy-Diffusion                       |                |               |
| D3.2                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                 | M30               | Benchmarking Report: Trainability, Temporal Expressibility & Noise Resilience |                |               |

|                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                       |               |                |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------|---------------|----------------|---------------|
| WP 4                                                                                                                                                                                                                                                                                                                                                                                  | High-Complexity Domain Validation (HEP, Neuro&Cyber)                                                                                                                                                                                                                                                                                                                                                                 |                   |                                                                       |               | Start month: 1 | End month: 36 |
| Contribution of project partners                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                       |               |                |               |
| Partner number                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                      | 1. SNU            | 2. Yonsei                                                             | 3. Fraunhofer | 4. Naples      |               |
| Total effort per partner (person*months)                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                      | 42                | 24                                                                    | 6             | 6              |               |
| <b>Aim of the WP</b><br>To validate the foundational breakthroughs (WP2/WP3) using massive, complex datasets from High Energy Physics (LHC), Neuroscience (EEG, fMRI), and Cybersecurity (network intrusion detection), proving quantum advantage in scalability, temporal modeling, temporal modeling, and certified robustness. This WP addresses <b>Objective 5 (Validation)</b> . |                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                       |               |                |               |
| Tasks                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                                       |               |                |               |
| T4.1                                                                                                                                                                                                                                                                                                                                                                                  | <b>Dataset Curation (M1–M12; Yonsei; SNU, Fraunhofer)</b><br>Preparation of LHC data including CMS calorimeter images, high-level physics features, and detector waveform signals (Yonsei); curation of EEG/fMRI time-series datasets (SNU); preparation of network intrusion detection datasets with PII-screened logs (Fraunhofer); implementation of quantum feature maps for high-dimensional data (Fraunhofer). |                   |                                                                       |               |                |               |
| T4.2                                                                                                                                                                                                                                                                                                                                                                                  | <b>Quantum Feature Mapping (M7–M18; Yonsei; SNU, Naples, Fraunhofer)</b><br>Development of quantum encoding schemes for HEP, Neuroscience, and Cybersecurity data; optimization of feature map architectures for domain-specific characteristics; implementation of dimension reduction strategies compatible with Multi-Chip protocols.                                                                             |                   |                                                                       |               |                |               |
| T4.3                                                                                                                                                                                                                                                                                                                                                                                  | <b>HEP Validation (Multi-Chip + LHC) (M13–M30; Yonsei; SNU, Fraunhofer)</b><br>Application of Multi-Chip Ensembles (from WP2) to LHC data classification using Quantum Vision Transformers; implementation of Q-ABCDiCo for background estimation; TCN-VQC for waveform denoising; benchmarking accuracy and generalization vs. single-chip and classical SOTA baselines.                                            |                   |                                                                       |               |                |               |
| T4.4                                                                                                                                                                                                                                                                                                                                                                                  | <b>Neuro Validation (Q-SSM + EEG/fMRI) (M13–M30; SNU; Naples, Fraunhofer)</b><br>Application of Q-SSM (from WP3) to EEG/fMRI spatio-temporal analysis; implementation of QFF-HQGA driven QAOA/VQE for brain network modularity optimization; evaluation of long-range temporal dependency capture vs. classical transformers.                                                                                        |                   |                                                                       |               |                |               |
| T4.5                                                                                                                                                                                                                                                                                                                                                                                  | <b>Cyber Validation (Fuzzy-Quantum Diffusion) (M13-M30; Fraunhofer; Naples, SNU)</b><br>Application of Fuzzy-Quantum Diffusion (from WP3) to network anomaly detection via hardware fingerprinting; QUARK-certified robustness testing against adversarial attacks; benchmarking detection rates and false positive rates vs. classical intrusion detection systems.                                                 |                   |                                                                       |               |                |               |
| T4.6                                                                                                                                                                                                                                                                                                                                                                                  | <b>Final Benchmarking vs SOTA (M25–M36; Yonsei; SNU, Naples, Fraunhofer)</b><br>Comprehensive benchmarking of all quantum approaches against classical state-of-the-art across HEP, Neuro, and Cyber domains; statistical analysis of quantum advantage claims; preparation of validation reports.                                                                                                                   |                   |                                                                       |               |                |               |
| Deliverable                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                      | Month of delivery | Title of deliverable                                                  |               |                |               |
| D4.1                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                      | M6                | Curated "Challenge Datasets" (LHC, EEG/fMRI, Network logs)            |               |                |               |
| D4.2                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                      | M36               | Validation Report: Roadmap to Quantum Advantage in HEP, Neuro & Cyber |               |                |               |

|                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                          |                   |                                                    |               |                |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------|---------------|----------------|---------------|
| WP 5                                                                                                                                                                                                                                                                 | Reliability, Safety & Certification                                                                                                                                                                                                                                                                                      |                   |                                                    |               | Start month: 1 | End month: 36 |
| Contribution of project partners                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                          |                   |                                                    |               |                |               |
| Partner number                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                          | 1. SNU            | 2. Yonsei                                          | 3. Fraunhofer | 4. Naples      |               |
| Total effort per partner (person*months)                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                          | 12                | 3                                                  | 15            | 9              |               |
| Aim of the WP                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                          |                   |                                                    |               |                |               |
| To define the first "QML Reliability Standard" and benchmark and certify the robustness of the developed QML models against cybersecurity threats, adversarial attacks, and privacy risks. This WP addresses <b>Objective 4 (Reliability)</b> certification aspects. |                                                                                                                                                                                                                                                                                                                          |                   |                                                    |               |                |               |
| Tasks                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                          |                   |                                                    |               |                |               |
| T5.1                                                                                                                                                                                                                                                                 | Requirements Specification (M1–M12; Fraunhofer; All)                                                                                                                                                                                                                                                                     |                   |                                                    |               |                |               |
|                                                                                                                                                                                                                                                                      | Definition and development of reliability and safety requirements for QML models; specification of robustness and reliability metrics (e.g., Lipschitz continuity, noise stability, data leakage); identification of cybersecurity vulnerability and inherent defenses relevant to model types developed in WP2 and WP3. |                   |                                                    |               |                |               |
| T5.2                                                                                                                                                                                                                                                                 | Robustness Enhancement (M7–M18; Fraunhofer; Naples)                                                                                                                                                                                                                                                                      |                   |                                                    |               |                |               |
|                                                                                                                                                                                                                                                                      | Development of cybersecurity defense mechanisms to improve model robustness, via architectural changes (e.g., robust feature maps, in cooperation with T4.2) and additional techniques (e.g., adversarial training, differential privacy).                                                                               |                   |                                                    |               |                |               |
| T5.3                                                                                                                                                                                                                                                                 | Vulnerabilities and Defenses Testing (M13–M30; Fraunhofer; Naples)                                                                                                                                                                                                                                                       |                   |                                                    |               |                |               |
|                                                                                                                                                                                                                                                                      | Application of Evolutionary QML to Network Intrusion Detection; stress-testing the Fuzzy-Quantum Diffusion models with gradient-based adversarial attacks; adversarial certification for recurrent (Q-SSM) and attention-based models.                                                                                   |                   |                                                    |               |                |               |
| T5.4                                                                                                                                                                                                                                                                 | Certification Protocol Development (M25–M36; Fraunhofer)                                                                                                                                                                                                                                                                 |                   |                                                    |               |                |               |
|                                                                                                                                                                                                                                                                      | Synthesis of results into a formal certification protocol for industrial QML applications; development of certification guidelines for safety-critical deployment; documentation of reliability standards.                                                                                                               |                   |                                                    |               |                |               |
| Deliverable                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                          | Month of delivery | Title of deliverable                               |               |                |               |
| D5.1                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                          | M18               | Robustness Report: Metrics & Initial Assessment    |               |                |               |
| D5.2                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                          | M36               | QML Reliability Standard & Certification Protocols |               |                |               |

|                                                                                                                                                          |                                             |                                                                                                                                                                                                                                                                                       |           |                                                 |                |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------|----------------|---------------|
| WP 6                                                                                                                                                     | Dissemination, Exploitation & Communication |                                                                                                                                                                                                                                                                                       |           |                                                 | Start month: 1 | End month: 36 |
| Contribution of project partners                                                                                                                         |                                             |                                                                                                                                                                                                                                                                                       |           |                                                 |                |               |
| Partner number                                                                                                                                           |                                             | 1. SNU                                                                                                                                                                                                                                                                                | 2. Yonsei | 3. Fraunhofer                                   | 4. Naples      |               |
| Total effort per partner (person*months)                                                                                                                 |                                             | 12                                                                                                                                                                                                                                                                                    | 4         | 5                                               | 5              |               |
| Aim of the WP                                                                                                                                            |                                             |                                                                                                                                                                                                                                                                                       |           |                                                 |                |               |
| To maximize impact through Open Science publications, release of the "Physics-Aware QML" library, and engagement with the industrial and public sectors. |                                             |                                                                                                                                                                                                                                                                                       |           |                                                 |                |               |
| Tasks                                                                                                                                                    |                                             |                                                                                                                                                                                                                                                                                       |           |                                                 |                |               |
| T6.1                                                                                                                                                     |                                             | Communication Planning (M1–M12; Fraunhofer; SNU)<br>Development of project website and communication strategy; establishment of social media presence; creation of project visual identity and materials.                                                                             |           |                                                 |                |               |
| T6.2                                                                                                                                                     |                                             | Publications & Conferences (M13–M36; All)<br>Publication of results in Open Access journals (Nature Comms, PRX Quantum, Quantum Machine Intelligence, etc.); management of the GitHub repository and Zenodo data archives; presentations at major conferences (IEEE QCE, CHEP, etc.). |           |                                                 |                |               |
| T6.3                                                                                                                                                     |                                             | Hackathon & Summer School (M25–M30; Fraunhofer; SNU)<br>Organization of the "Quantum Robustness Hackathon" (M30) challenging participants to attack/defend QML models; summer school on Physics-Aware QML for early-career researchers.                                               |           |                                                 |                |               |
| T6.4                                                                                                                                                     |                                             | Open-Source Library Release (M31–M36; SNU; Naples, Fraunhofer)<br>Final release of the integrated "Physics-Aware QML" open-source library under Apache 2.0 license; comprehensive documentation and Jupyter notebook tutorials; Zenodo archival with DOI assignment.                  |           |                                                 |                |               |
| Deliverable                                                                                                                                              |                                             | Month of delivery                                                                                                                                                                                                                                                                     |           | Title of deliverable                            |                |               |
| D6.1                                                                                                                                                     |                                             | M36                                                                                                                                                                                                                                                                                   |           | Open Source "Physics-Aware QML" Library Release |                |               |
| D6.2                                                                                                                                                     |                                             | M36                                                                                                                                                                                                                                                                                   |           | Final Plan for Exploitation and Standardization |                |               |

#### Work package overview (total effort per WP and partner in person.months)

| Partner       | WP1       | WP2       | WP3       | WP4       | WP5       | WP6       | Total      |
|---------------|-----------|-----------|-----------|-----------|-----------|-----------|------------|
| 1. SNU        | 9         | 60        | 36        | 42        | 12        | 12        | 171        |
| 2. Yonsei     | 1         | 6         | 2         | 24        | 3         | 4         | 40         |
| 3. Fraunhofer | 2         | 4         | 4         | 6         | 15        | 5         | 36         |
| 4. Naples     | 1         | 9         | 24        | 6         | 9         | 5         | 54         |
| <b>Total</b>  | <b>13</b> | <b>79</b> | <b>66</b> | <b>78</b> | <b>39</b> | <b>26</b> | <b>301</b> |

### 3.3 Management structure, milestones, risk assessment

#### Organizational Structure and Decision-Making

**Project Coordinator (PC):** Prof. Jiook Cha (SNU) is the single point of contact for the QuantERA Secretariat. The PC is responsible for overall progress monitoring, submission of deliverables, and conflict resolution.

**General Assembly (GA):** The ultimate decision-making body, composed of one representative from each partner. The GA meets bi-annually to review progress against Milestones. Decisions are made by consensus; if consensus fails, a majority vote (one vote per partner) applies.

**Scientific Steering Committee (SSC):** Composed of the Work Package Leaders (WP1/WP2: SNU; WP3: Naples; WP4: Yonsei; WP5/WP6: Fraunhofer). The SSC meets quarterly to ensure technical alignment across WPs.

**Innovation & Ethics Board:** Led by Fraunhofer IKS, this board oversees IPR management, Open Science compliance, and ethical handling of sensitive data (Neuroscience EEG/fMRI, Cybersecurity network logs).

#### Appropriateness of Structure

This structure is appropriate because it is **lean** (minimizing administrative overhead for a 4-partner consortium) yet **robust** (ensuring all scientific disciplines—Physics, Computer Science, Engineering—have equal representation in the GA). The geographic distribution is balanced with clear communication channels to bridge the Asia-Europe collaboration.

#### List of Milestones

| Milestone | Month | WP Involved        | Title                               |
|-----------|-------|--------------------|-------------------------------------|
| MS1       | M12   | WP2, WP3, WP4, WP5 | Theoretical Foundations Established |
| MS2       | M24   | WP2, WP3, WP4      | Algorithmic Validation (Simulation) |
| MS3       | M30   | WP4, WP5           | Cross-Domain Integration            |
| MS4       | M36   | WP2, WP3, WP5, WP6 | Final Deliverables                  |

#### Success Criteria

- **MS1:** Multi-Chip Ensemble protocol specifications complete; QFF-HQGA and Q-SSM architectures defined; Fuzzy Logic Controllers specified; reliability metrics defined; Domain datasets (LHC, EEG/fMRI) preprocessed and feature-mapped
- **MS2:** QFF-HQGA converges on >10-layer circuits where backpropagation fails; Q-SSM achieves  $\geq 1.5\times$  classical memory capacity; Fuzzy-Quantum Diffusion achieves FID  $\leq 1.5\times$  classical diffusion baseline
- **MS3:** Multi-Chip Ensemble reconstructs LHC particle correlations across 2+ parallel chips; Q-SSM demonstrates superior long-range temporal capture on EEG/fMRI; QML models pass robustness certification against adversarial attacks
- **MS4:** Open-source "Physics-Aware QML" library released (Apache 2.0); Final benchmarks published demonstrating advantage in scalability (O1), trainability (O2), temporal expressibility (O3), and reliability (O4: Lipschitz  $L < 2.0$  certification); Certification Standard documented

## Critical Risks for Implementation

| Description of Risk                                                                                                              | WPs      | Proposed Risk-Mitigation Measures                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scientific:</b> Barren Plateaus persist despite QFF-HQGA methods, preventing convergence in deep circuits.                    | WP3, WP4 | <b>Mitigation:</b> We do not rely on a single optimization strategy. The HQGA component provides gradient-free global search via entangled crossover and quantum elitism. If QFF proves unstable, HQGA serves as a robust fallback that navigates the optimization landscape without gradient dependence. Additionally, the local "goodness" objectives of QFF inherently reduce gradient variance.          |
| <b>Technical:</b> Ensemble Aggregation fails to achieve accuracy targets due to weak individual learners on small chips.         | WP2, WP4 | <b>Mitigation:</b> We will implement advanced "Weighted Voting" mechanisms where each chip's output weight is determined by its reliability score. The Quantum Transformer ansatz is specifically designed to maximize expressibility even on small chips. Selective Entanglement between chips processing globally dependent features will enhance collective expressivity.                                 |
| <b>Technical:</b> Q-SSM fails to capture long-range temporal correlations in EEG/fMRI data, underperforming classical baselines. | WP3, WP4 | <b>Mitigation:</b> The three-branch quantum superposition ( $\alpha, \beta, \gamma \in \mathbb{C}$ ) architecture provides interference-like effects while LSTM-style gating handles sequential memory. If temporal modeling is insufficient, we will enhance the bidirectional processing layer and increase chunk overlap. Alternative QSVT-enhanced attention mechanisms provide a fallback architecture. |
| <b>Technical:</b> Fuzzy-Quantum Diffusion fails to achieve target generative fidelity under NISQ noise conditions.               | WP3, WP4 | <b>Mitigation:</b> The Fuzzy Logic Controllers using POVMs (Positive Operator-Valued Measures) are specifically designed to model noise as continuous "degrees of uncertainty" rather than binary errors. If the Hardware-Attention U-Net underperforms, we can pivot to Quantum GANs enhanced with the same Fuzzy Controllers to stabilize adversarial training dynamics.                                   |
| <b>Operational:</b> Delays in accessing sensitive domain data (LHC physics data, EEG/fMRI clinical data, network security logs). | WP4, WP5 | <b>Mitigation:</b> All partners have established direct access: Prof. Yoo is a CMS Collaboration member with CERN LHC access; SNU has IRB-approved neuroimaging protocols. For contingency, open-source proxy datasets (CERN Open Data Portal, PhysioNet EEG archives, KDD Cup 99) are identified to proceed with algorithmic development without delay.                                                     |
| <b>Management:</b> IPR disputes regarding industrial exploitation of jointly-developed algorithms.                               | WP1, WP6 | <b>Mitigation:</b> A comprehensive Consortium Agreement (CA) based on the DESCA model will be signed before project start, clearly defining background/foreground IP ownership and licensing terms. The "Open Source by Default" strategy (Apache 2.0) for the core library minimizes commercial disputes while preserving academic recognition.                                                             |

### 3.4 Consortium as a whole

#### Matching Objectives and Expertise

The consortium is designed as a **trans-continental feedback loop** bridging Foundational Quantum Theory (O1–O4) and Rigorous Domain Validation (O5). No single partner possesses the full spectrum of skills required to deliver the "Physics-Aware QML Stack," necessitating this specific four-partner collaboration:

| Role       | Partner                  | Objective Addressed               | Key Contribution                                                |
|------------|--------------------------|-----------------------------------|-----------------------------------------------------------------|
| Architects | SNU (Korea)              | O1: Scalability                   | Multi-Chip Ensemble protocols, Q-SSM, Quantum Transformers      |
| Optimizers | Naples (Italy)           | O2: Trainability, O4: Reliability | QFF-HQGA, Fuzzy Quantum Diffusion, POVMs                        |
| Validators | Yonsei (Korea)           | O5: Domain Validation             | LHC/CMS data, Q-ABCDISCo, TCN-VQC, Quantum ViT                  |
| Certifiers | Fraunhofer IKS (Germany) | O4: Reliability Certification     | Robust QML, Lipschitz bounds, adversarial testing, benchmarking |

#### Complementarity and Contribution

The partners complement each other through a "**Scale-Logic-Trust**" triangle with domain grounding:

- **SNU (Scale)** Contributes **171 PM** (57% of total effort), leading WP1/WP2. Provides scalable architectures (Multi-Chip Ensembles with Selective Entanglement, variance reduction  $\propto 1/k$ ) essential for fragmented NISQ hardware.
- **Naples (Logic)** Contributes **54 PM** leading WP3. Without Naples, deep architectures would be untrainable (Barren Plateaus). QFF-HQGA (entangled crossover, quantum elitism) and Fuzzy Logic Controllers using POVMs provide the resilience layer for hardware-native generative modeling.
- **Fraunhofer IKS (Trust)** Contributes **36 PM** leading WP5/WP6. Transforms outputs into certified software assets (Lipschitz  $L < 2.0$  guarantees) via robustness analyses, enabling safety-critical deployment.
- **Yonsei (Physical Reality)** Contributes **40 PM** leading WP4. CMS Collaboration membership provides LHC petabyte-scale data with intrinsic non-local correlations—the ideal testbed proving Multi-Chip Ensembles preserve quantum expressivity.

**Resource Adequacy:** The **301 Person-Months** ensure adequate capacity: SNU's large team handles coordination (WP1) and architecture development (WP2); Naples provides specialized optimization expertise; Fraunhofer brings industrial certification experience; Yonsei manages the HEP data pipeline.

#### Industrial/Commercial Involvement

Fraunhofer IKS bridges foundational research to industry. As a leading German RTO coordinating one of the **Munich Quantum Valley** consortia and European QC Benchmarking Committee, Fraunhofer provides direct channels to automotive, aerospace, and medical technology sectors.

This industrial involvement is consistent with Section 2 exploitation measures: The "**QML Reliability Standard**" (WP5, D5.2) validates commercial viability of "Trustworthy QML" services with formal metrics (gradient variance, entangling capability, Lipschitz bounds). The "**Open Source by Default**" strategy (Apache 2.0) maximizes academic impact while the certification layer creates commercial value through consulting services for companies requiring safety guarantees before QML deployment.

### 3.5 Description of the consortium

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Jiook Cha</b><br>Project Coordinator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Seoul National University / Department of Psychology,<br/>Department of Brain and Cognitive Sciences,<br/>Interdisciplinary Program in Artificial Intelligence</b> |
| <b>Expertise:</b><br><br>Seoul National University is South Korea's premier research university. Prof. Jiook Cha's lab specializes in <b>Quantum Machine Learning (QML) architectures</b> and their application to <b>Computational Neuroscience</b> . The group possesses unique expertise in:<br><br><ol style="list-style-type: none"><li>1. <b>Multi-Chip Ensembles:</b> Developing scalable approaches to train QML models across multiple chips, applied to image data, EEG/fMRI, and Reinforcement Learning.</li><li>2. <b>Quantum Time-Series Analysis:</b> Pioneering quantum algorithms for temporal data, including Quantum Transformers and Quantum State Space Models (Q-SSM) for fMRI/EEG analysis.</li><li>3. <b>QML for Neuroscience:</b> Bridging quantum computing with brain science through specialized neural network architectures including Quantum Time-series Transformers and Hybrid Quantum Temporal Convolutional Networks.</li></ol><br><b>Principal Investigator: Prof. Jiook Cha (Male)</b><br><br>Prof. Cha is a leading expert in applied QML architectures and computational neuroscience. His research focuses on creating scalable quantum algorithms for high-dimensional biological data, with particular emphasis on overcoming NISQ hardware limitations through distributed computing strategies.<br><br><b>Relevant Achievements:</b> <ul style="list-style-type: none"><li>• Developing <b>multi-chip ensembles</b> approaches and applying them to image data, EEG data, and reinforcement learning environments.</li><li>• Developing <b>quantum time-series transformers</b> and applying them to fMRI analysis.</li><li>• Developing <b>hybrid quantum temporal convolutional neural network (HQTCN)</b> algorithms and applying them to EEG analysis.</li></ul> |                                                                                                                                                                       |
| <b>Role in project: Coordinator &amp; QML Architecture Lead.</b><br><br>SNU leads WP1 (Management) and drives WP2 (Scalable Architectures) by implementing the Multi-Chip Ensemble protocols. Prof. Cha also leads the theoretical development of the Neuro-Generative validation in WP4, utilizing his expertise in Quantum Transformers and HQTCNs to process EEG/fMRI data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                       |



| Hwidong Yoo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Yonsei University / Department of Physics |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <p><b>Expertise:</b></p> <p>Yonsei University is a leading private research university in Korea with a strong legacy in High Energy Physics (HEP). Prof. Hwidong Yoo is a key member of the <b>CMS Collaboration at CERN</b>, providing the consortium with direct access to massive-scale "Big Science" data. Yonsei hosts <b>IBM Quantum System One (127-qubit)</b>, deployed November 2024—the second such system on a university campus worldwide.</p> <ol style="list-style-type: none"> <li>1. <b>High Energy Physics:</b> Deep expertise in LHC particle jet analysis, tracking, event classification, and detector physics. Key contributor to the <b>Snowmass 2021 White Paper</b> on dual-readout calorimetry shaping future detector R&amp;D.</li> <li>2. <b>Dual-Readout Calorimetry:</b> Pioneering work on 3D shower shape reconstruction using SiPM timing, producing <b>5-dimensional spatio-temporal tensors (x, y, z, E, t)</b>—the exact data structure motivating the Quantum Time-series Transformer.</li> <li>3. <b>Scalable Computing:</b> Experience scaling deep learning on <b>HPC infrastructure (Nurion supercomputer at KISTI)</b> across thousands of parallel processors, directly applicable to distributed quantum system coordination.</li> <li>4. <b>Quantum for HEP:</b> Experience in benchmarking quantum algorithms (Quantum Vision Transformers, Q-ABCDiCo, TCN-VQC) against classical Deep Learning for particle physics tasks.</li> </ol> <p><b>Principal Investigator: Prof. Hwidong Yoo (Male)</b></p> <p>Prof. Yoo is an expert in experimental high-energy physics and the application of advanced computing to particle detection. He leads the validation efforts, bridging the gap between abstract quantum algorithms and physical reality.</p> <p><b>Relevant Achievements:</b></p> <ul style="list-style-type: none"> <li>• Leadership roles in the <b>CMS Collaboration (CERN)</b> for particle data analysis; key author of Snowmass 2021 dual-readout calorimetry white paper.</li> <li>• Development of classical baselines for CMS calorimeter image, high-level feature, and waveform analysis tasks.</li> <li>• <b>Large-scale distributed training</b> of CNNs on Nurion HPC (1.85× signal efficiency improvement over cut-based methods).</li> <li>• Pioneering work in applying <b>Quantum Vision Transformers, Quantum Neural Networks (Q-ABCDiCo)</b>, and <b>hybrid TCN-VQC models</b> to CMS classification, background estimation, and waveform denoising.</li> </ul> |                                           |
| <p><b>Role in project: Domain Validation Lead (HEP).</b></p> <p>Yonsei leads <b>WP4 (High-Complexity Domain Validation)</b>. They curate LHC datasets (CMS calorimeter images, high-level physics features, detector waveforms), implement quantum feature maps for high-dimensional particle data, and rigorously benchmark Multi-Chip Ensemble protocols (WP2) against classical SOTA. Prof. Yoo's HPC scaling experience directly supports <b>WP2 (Scalable Distributed Quantum Architectures)</b>, while Yonsei's IBM Quantum System One provides critical hardware validation infrastructure.</p> <p><b>Total effort: 40 PM.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                           |

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| Jeanette Miriam Lorenz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Fraunhofer Institute for Cognitive Systems IKS /<br>Department "Quantum-enhanced AI" |
| <p><b>Expertise:</b><br/> Fraunhofer IKS is Germany's leading applied research institute for <b>Safe and Reliable Software Engineering and Artificial Intelligence ("Safe Intelligence")</b>. PD Dr. habil. Jeanette Miriam Lorenz leads the <b>QuaST project</b> (quantum-enabling services and tools for industrial applications) and is affiliated with the <b>Munich Quantum Valley (MQV)</b> and <b>Bavarian Competence Center for Quantum Security and Data Science (BayQS)</b>. Her group specializes in <b>Reliable Quantum Computing for Applications</b>, uniquely combining quantum generalization theory with industrial-scale benchmarking frameworks.</p> <ul style="list-style-type: none"> <li>- <b>Quantum Generalization Theory:</b> Pioneering <b>generalization bounds for hybrid quantum-classical models</b> (2025), proving distributed architectures maintain performance—crucial for validating Multi-Chip Ensembles (WP2).</li> <li>- <b>Quantum Randomized Smoothing:</b> Demonstrated <b>quadratic speedups</b> in robustness certification using Quantum Amplitude Estimation (2024), applicable to Prof. Yoo's HEP 5D tensor data where classical certification is prohibitive.</li> <li>- <b>Noise-as-Resource Theory:</b> Theoretical grounding (2024) for repurposing quantum noise in diffusion processes—directly informing Fuzzy-Quantum Diffusion collaboration with Naples (WP3).</li> <li>- <b>Industrial Benchmarking &amp; QUARK Framework:</b> Developer of the open-source <b>QUARK</b> benchmarking platform for systematic QML evaluation including noise resilience and scalability testing.</li> </ul> <p><b>Principal Investigator: PD Dr. habil. Jeanette Miriam Lorenz (Female)</b><br/> PD Dr. Lorenz heads the department 'Quantum-enhanced AI' at Fraunhofer IKS and is associate professor at <b>LMU Munich</b> and brings extensive <b>ATLAS experiment (CERN LHC)</b> experience in supersymmetry searches, dark matter, and Higgs physics. Her research establishes <b>rigorous theoretical guarantees</b> of QML performance beyond heuristic claims.</p> <p><b>Relevant Achievements:</b></p> <ul style="list-style-type: none"> <li>• <b>Generalization bounds for hybrid QML</b> (2025), proving shallow evolutionary-trained circuits match deeper gradient-optimized circuits.</li> <li>• <b>Quantum Randomized Smoothing</b> achieving quadratic certification speedups via Grover-based amplitude estimation (2024).</li> <li>• <b>QUARK framework</b> leadership; <b>quantum-classical CNNs for medical imaging</b> outperforming classical baselines with limited training data (IEEE QCE, 2022).</li> <li>• <b>Quantum-classical CNNs for medical imaging</b> (IEEE QCE, 2022) showing quantum models outperform classical baselines on breast tumor recognition, particularly with limited training data.</li> <li>• Coordination of the <b>European QC Benchmarking Coordination Committee (EQCBC)</b>.</li> </ul> |                                                                                      |
| <p><b>Role in project: Reliability &amp; Certification Lead</b><br/> Fraunhofer leads <b>WP5 (Reliability, Safety &amp; Certification)</b> and <b>WP6 (Dissemination, Exploitation &amp; Communication)</b>. Dr. Lorenz provides <b>QML Reliability Standard</b> ensuring unified benchmarking across all partners. Her generalization theory validates Multi-Chip Ensembles (WP2); noise-as-resource research informs Fuzzy-Quantum Diffusion (WP3); Quantum Randomized Smoothing certifies Q-SSM and attention models for HEP time-series (WP4). Fraunhofer leads the Cybersecurity domain validation (T4.5), applying Fuzzy-Quantum Diffusion and QML Reliability Standards to network intrusion detection. Her ATLAS/CERN statistical expertise supports HEP and Neuro validation. She</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                      |

coordinates the "Quantum Robustness Hackathon" and Open Science compliance. **Total effort: 36 PM.**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                           |
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| Giovanni Acampora                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | University of Naples Federico II / Department of Physics “Ettore Pancini” |
| <p><b>Expertise:</b></p> <p>The University of Naples Federico II is one of the oldest universities in the world. Prof. Giovanni Acampora's <b>QUASAR Lab</b> (Quantum Computing and Smart Systems Laboratory) is globally recognized for <b>Computational Intelligence</b> and <b>Fuzzy Logic</b>. They bring the critical "classical intelligence" layer to the quantum stack.</p> <ol style="list-style-type: none"><li>1. <b>Fuzzy Logic Systems:</b> World-class expertise in Type-2 Fuzzy Sets and Fuzzy Logic Controllers, essential for modeling NISQ noise as continuous uncertainty.</li><li>2. <b>Evolutionary Computation:</b> Deep experience in Hybrid Quantum Genetic Algorithms (HQGA) and bio-inspired optimization to solve non-convex problems; developer of <b>EVOVAQ</b>, the first open-source evolutionary algorithm toolbox for variational quantum circuits (SoftwareX, 2024).</li><li>3. <b>Quantum Computational Intelligence:</b> Pioneering the fusion of Fuzzy Logic with Quantum Computing, including Positive Operator-Valued Measures (POVMs) for "unsharp" quantum measurements.</li><li>4. <b>Distributed Quantum Systems:</b> Demonstrated <b>quantum fuzzy inference engines on the D-NISQ model</b>, proving divide-and-conquer approaches viable for multi-chip quantum architectures.</li></ol> <p><b>Principal Investigator: Prof. Giovanni Acampora (Male)</b></p> <p>Prof. Acampora is a Senior Member of IEEE, <b>Editor-in-Chief of Springer Quantum Machine Intelligence</b>, and a pioneer in Quantum Computational Intelligence. His work focuses on hybridizing soft computing with quantum mechanics to solve trainability and noise-resilience issues.</p> <p><b>Relevant Achievements:</b></p> <ul style="list-style-type: none"><li>• Definition of <b>Quantum Fuzzy Logic</b> frameworks for uncertainty management using POVMs.</li><li>• Development of <b>Hybrid Quantum Genetic Algorithms (HQGA)</b> with entangled crossover and quantum elitism for optimizing Variational Quantum Circuits.</li><li>• Application of <b>Quantum Forward-Forward (QFF)</b> algorithms with local "goodness" objectives to prevent Barren Plateaus in QML.</li><li>• Hardware validation of <b>quantum fuzzy inference engines</b> on both superconducting and ion-trap backends, demonstrating real-world implementability (IEEE TQE, 2024).</li><li>• Analysis of <b>quantum noise effects on evolutionary algorithms</b>, providing theoretical grounding for the noise-as-resource paradigm.</li></ul> <p><b>Role in project: Optimization &amp; Intelligence Lead</b></p> <p>Naples leads <b>WP3 (Physics-Informed Optimization &amp; Generative Models)</b>. They provide the QFF-HQGA algorithms to drive gradient-free training, design the Fuzzy Logic Controllers for the Fuzzy-Quantum Diffusion models using POVMs, and develop Hardware-Attention U-Net architectures for noise-resilient generative modeling. Prof. Acampora's EVOVAQ framework provides the software infrastructure for <b>WP2 (Scalable Distributed Quantum Architectures)</b>, while his D-NISQ expertise validates the divide-and-conquer approach. Naples also supports <b>WP4 (Neuro Validation)</b> through</p> |                                                                           |

QFF-HQGA driven brain connectivity optimization, and **WP5** in adversarial evolutionary testing. **Total effort: 54 PM.**

### 3.6 Consortium agreement principles (partner's rights and duties, IPR management, data management, open access, GDPR)

(max. ½ page)

#### Consortium Agreement & Decision Making

The consortium will sign a Consortium Agreement (CA) based on the DESCA (Horizon Europe) model prior to the project start. The CA will define the rights and duties of partners, specifying that SNU (Coordinator) manages the administrative interface, while the General Assembly (one vote per partner) decides on major scientific pivots.

#### Intellectual Property Rights (IPR) Management

- **Background IP:** Partners retain ownership of their pre-existing know-how. Specifically, **Univ. Naples** retains rights to their pre-existing Fuzzy Logic libraries.
- **Foreground IP:** The new "Physics-Aware QML" algorithms (e.g., Multi-Chip Ensemble protocols, Fuzzy-Diffusion architectures) developed jointly will be jointly owned. We adopt an "**Open Source by Default**" strategy for software outputs, releasing the unified library under an **Apache 2.0 license** to maximize adoption.
- **Commercial Exploitation:** Fraunhofer IKS leads the exploitation strategy. If a patentable invention arises (e.g., a specific industrial control system based on Robust QML), the CA outlines fair access rights and royalty-sharing mechanisms for the inventors.

#### Data Management & Open Access

- **FAIR Data:** We adhere to the **FAIR principles** (Findable, Accessible, Interoperable, Reusable). All non-sensitive "Challenge Datasets" (LHC jets, synthetic EEG) will be deposited in **Zenodo** with DOIs.
- **Open Access:** The consortium commits to **100% Open Access** for peer-reviewed publications (Green or Gold), ensuring immediate availability on **arXiv**.

#### GDPR & Ethical Compliance

- **Neuroscience Data:** The processing of EEG/fMRI data (WP4) involves human subjects. **SNU** will ensure all data is **pseudonymized** at the source before sharing within the consortium, in strict compliance with **GDPR** and Korean Personal Information Protection Act (PIPA).
- **Cybersecurity Data:** Network logs used in WP5 will be screened to ensure no personally identifiable information (PII) or critical infrastructure vulnerabilities are exposed.

### 3.7 Significant facilities and large equipment available to the consortium to perform the project

(max. ½ page)

The consortium brings together world-class infrastructure that enables "Big Science" validation without the need for external hardware procurement.

### 1. High-Performance Computing & Simulation (SNU & Naples)

- **SNU:** Access to extensive GPU clusters required to simulate the **Multi-Chip Ensemble** protocols. Since we are simulating distributed quantum learning (validating >20 qubits across multiple virtual chips), high-memory classical simulation resources are critical.
- **Univ. Naples:** Dedicated computational clusters optimized for **Evolutionary Algorithms**, which require massive parallel processing capabilities to evolve the populations for the Memetic QFF optimization tasks.

### 2. "Big Science" Data Infrastructure (Yonsei)

- **Yonsei (CMS Collaboration):** Through Prof. Yoo's leadership in the CMS Collaboration, the consortium has authorized access to the **CERN LHC computing grid** and petabyte-scale storage systems. This provides the massive **High Energy Physics datasets** (particle jets) required for WP4, which are otherwise inaccessible to standard research groups.

### 3. Reliability Benchmarking & Quantum Hardware Access (Fraunhofer IKS)

- **The QUARK Framework:** Fraunhofer provides its expertise in the open-source **QUARK benchmarking suite**, a significant software facility that automates benchmarking. This pre-existing infrastructure can accelerate the development.
- **Munich Quantum Valley (MQV):** As a key member of MQV, Fraunhofer IKS has access to a diverse ecosystem of quantum hardware backends (Superconducting, Ion Trap) for validating the **Multi-Chip** protocols on physical NISQ devices where applicable.

## 3.8 Link with ongoing projects

(max. ½ page)

For each partner indicate (if applicable) the ongoing projects linked to the proposal topic, and their funding sources.

## 3.9 Financial plan

(max. 1 page)

The resources to be committed for each project partner have to be described in the Electronic Submission System by the coordinator. These resources include: Personnel, Consumables, Equipment, Travel, Subcontracting, Provisions, Licensing fees, other. Justify them here. Both the justification and the information in the system will be communicated to the Evaluation Panel.

The total requested budget for the consortium is **€1,065,945.34**, distributed to ensure the successful execution of the "Physics-Aware QML" work program.

### 1. Personnel Costs (€569,211.76)

This is the largest budget category, reflecting the labor-intensive nature of foundational algorithmic research.

- **Partner 1 (SNU):** Requesting **€163,905.88** to support full-time PhD students dedicated to developing the **Multi-Chip Ensemble** protocols (WP2) and **Quantum Transformers** (WP4).
- **Partner 2 (Yonsei):** Requesting **€84,705.88** for three graduate students to handle the massive data preprocessing and benchmarking of **LHC particle jets** (WP4).
- **Partner 3 (Naples):** Requesting **€56,000** for a Post-doc researcher specializing in **Evolutionary Algorithms** and **Fuzzy Logic** to lead the optimization tasks in WP3.
- **Partner 4 (Fraunhofer IKS):** Requesting **€264,600** for a senior researcher/Post-doc with industrial experience to drive the **certification** and **Reliability Standards** (WP5).

## 2. Equipment (€71,098.04)

Essential hardware is required for simulating quantum circuits and training hybrid models.

- **Partner 1 (SNU) & Partner 3 (Naples):** Requesting **€16,833.33** and **€12,500** respectively for **high-performance GPU workstations**. These are critical for simulating the **Quantum Forward-Forward (QFF)** algorithm, which requires intensive classical matrix operations during the "local goodness" updates (WP3).
- **Partner 2 (Yonsei):** Requesting **€11,764.71** for servers to store and process the petabyte-scale **LHC datasets** locally before quantum feature mapping (WP4).
- **Partner 4 (Fraunhofer IKS):** Requesting **€30,000** specifically for **"Access to Quantum Computers."** This budget is allocated to purchase compute time on cloud-based NISQ backends (e.g., IBM Quantum, IonQ) to validate the **Robustness metrics** on real hardware (WP5).

## 3. Travel (€162,122.36)

Travel costs are strictly necessary for ensuring cohesion between the Asian and European partners.

- **Partner 1 (SNU) & Partner 2 (Yonsei):** Requesting **~€70,000 each**. This higher amount is justified by the long-distance travel from South Korea to Europe for **bi-annual consortium meetings** and "Methodology Swap" workshops (WP6). It also covers travel for PhD students to attend major conferences (e.g., QIP, IEEE CEC) to disseminate results.
- **Partner 3 (Naples) & Partner 4 (Fraunhofer IKS):** Requesting **€7,000** and **€15,000** respectively for intra-European travel to consortium meetings and the final "Quantum Robustness Hackathon" in Munich, as well as for publications via international high-profile conferences.

#### 4. Other Goods, Services & Consumables (€56,104.71)

- **Publication Fees:** A significant portion (approx. **€3,500** per partner) is allocated for **Gold Open Access** publication fees to ensure immediate availability of our foundational results, complying with Open Science mandates.
- **Partner 2 (Yonsei):** Requesting **€11,764.71** for consumables related to data storage media and minor research supplies for the HEP validation experiments.

#### 5. Overheads (€207,408.47)

Calculated according to the specific national rules of each funding agency (NRF Korea, MUR Italy, DFG Germany).

#### Summary of Resource Commitment

The allocation ensures that SNU and Yonsei have the personnel and travel resources to bridge the geographical gap and drive the heavy theoretical/experimental load, while Fraunhofer IKS is adequately resourced to access real quantum hardware for industrial-grade certification. Univ. Naples is efficiently budgeted to provide the specialized optimization logic.

## 4. Ethical issues (max. ½ page)

Describe any foreseeable ethical issue that may arise during the course of the research project. Describe all mitigation strategies employed to reduce ethical risk, and justify the research methodology with respect to ethical issues.

### Foreseeable Ethical Issues

This project involves two sensitive domains that require rigorous ethical oversight:

1. **Human Biological Data (Neuroscience):** The processing of EEG and fMRI data (WP4) raises concerns regarding participant privacy, data protection (GDPR), and potential re-identification risks.
2. **Dual-Use Potential (Cybersecurity):** The development of adversarial attack protocols (WP5) and robust AI models involves "dual-use" technologies that, while designed for defense, could theoretically be misused to bypass existing security systems.

### Mitigation Strategies

We implement a "Safety-by-Design" approach to mitigate these risks:

- **Privacy-Enhancing Technologies (PETs):** For Neuroscience data, we strictly adhere to **GDPR** and the **Declaration of Helsinki**. All data will be pseudonymized at the source (SNU/Naples) before being processed by the quantum algorithms. Furthermore, a core scientific objective of this project is the development of **Fuzzy-Quantum Diffusion Models** (WP3) specifically to generate high-fidelity *synthetic* data. This methodology serves as a direct ethical mitigation, allowing us to share and process realistic brain data without exposing sensitive patient information.
- **Responsible Cybersecurity Research:** To mitigate dual-use risks, we strictly follow **Responsible Disclosure** guidelines. Detailed specifications of the adversarial attack vectors will be shared only within the consortium via encrypted channels ("As Closed As Necessary"). Public dissemination will focus on the *defense mechanisms* (Robustness Certification) rather than the attack tools.
- **Oversight:** An **Innovation & Ethics Board** (led by Fraunhofer IKS) will monitor compliance throughout the project lifecycle, ensuring that all AI models are tested for bias (e.g., sex-specific variances in EEG) before validation.

## Justification of Methodology

The use of high-dimensional human data is scientifically necessary to validate the Temporal Expressibility of our Quantum State Space Models (Q-SSM). Synthetic or toy datasets lack the complex, long-range correlations required to prove quantum advantage. Similarly, simulating adversarial attacks is the only mathematically rigorous way to certify the Reliability of QML models using the QUARK framework, a necessary step to ensure future quantum technologies are safe for society.

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